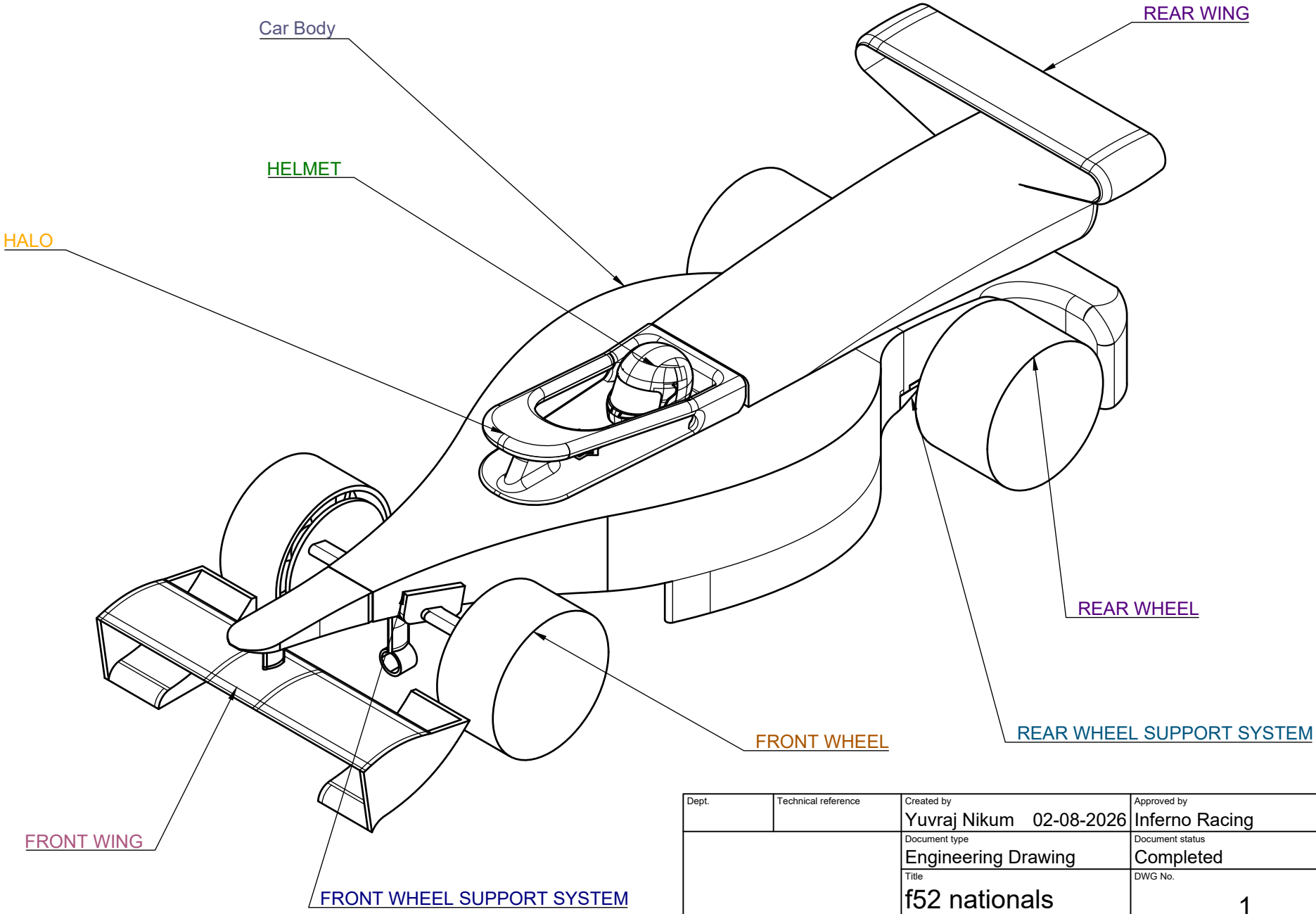
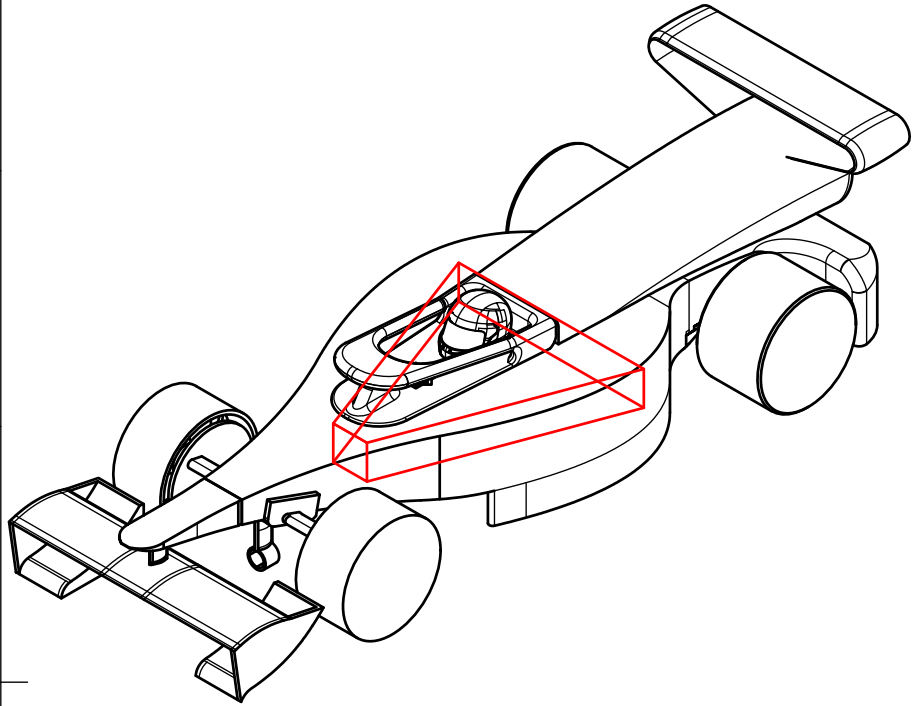
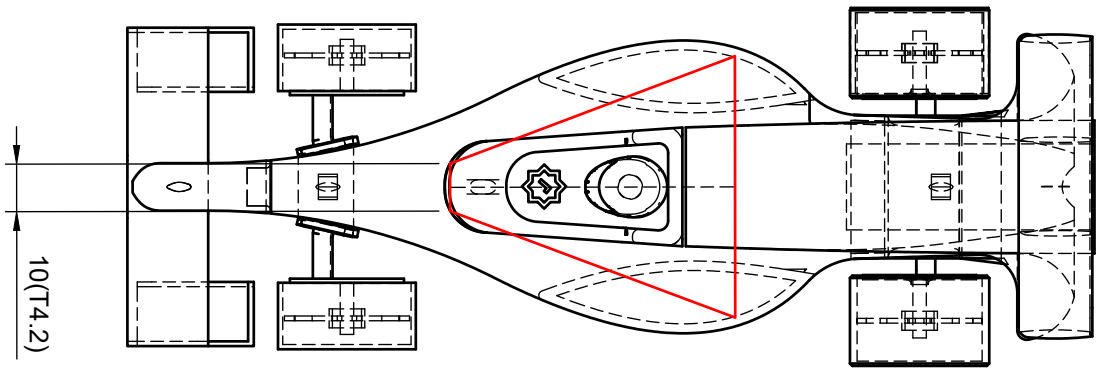
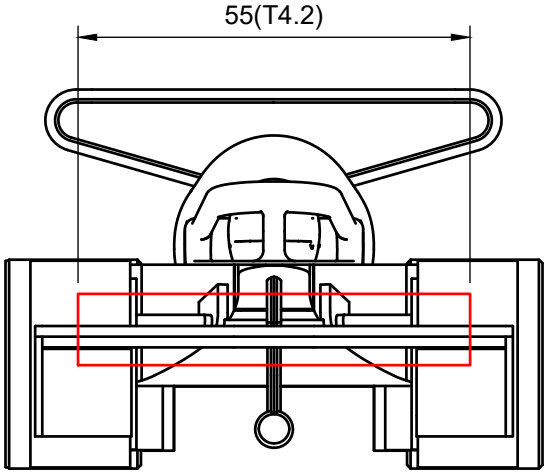
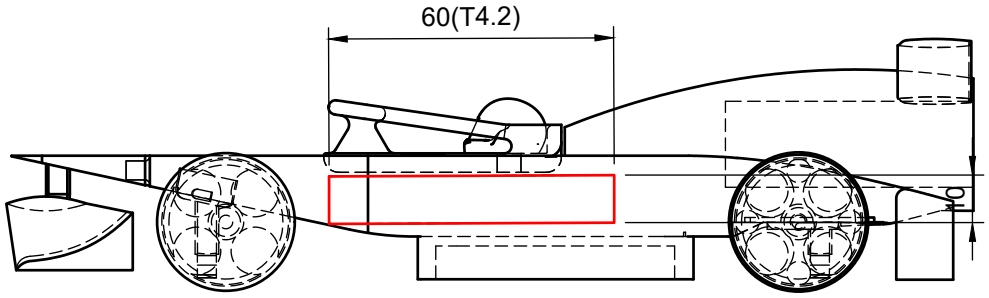



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| | | Yuvraj Nikum 02-08-2026 | Inferno Racing |
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| | | Title: f52 nationals | DWG No.: 1 |
| | | | Rev. / Date of issue: 01.08.2026 / Sheet: 1/10 |

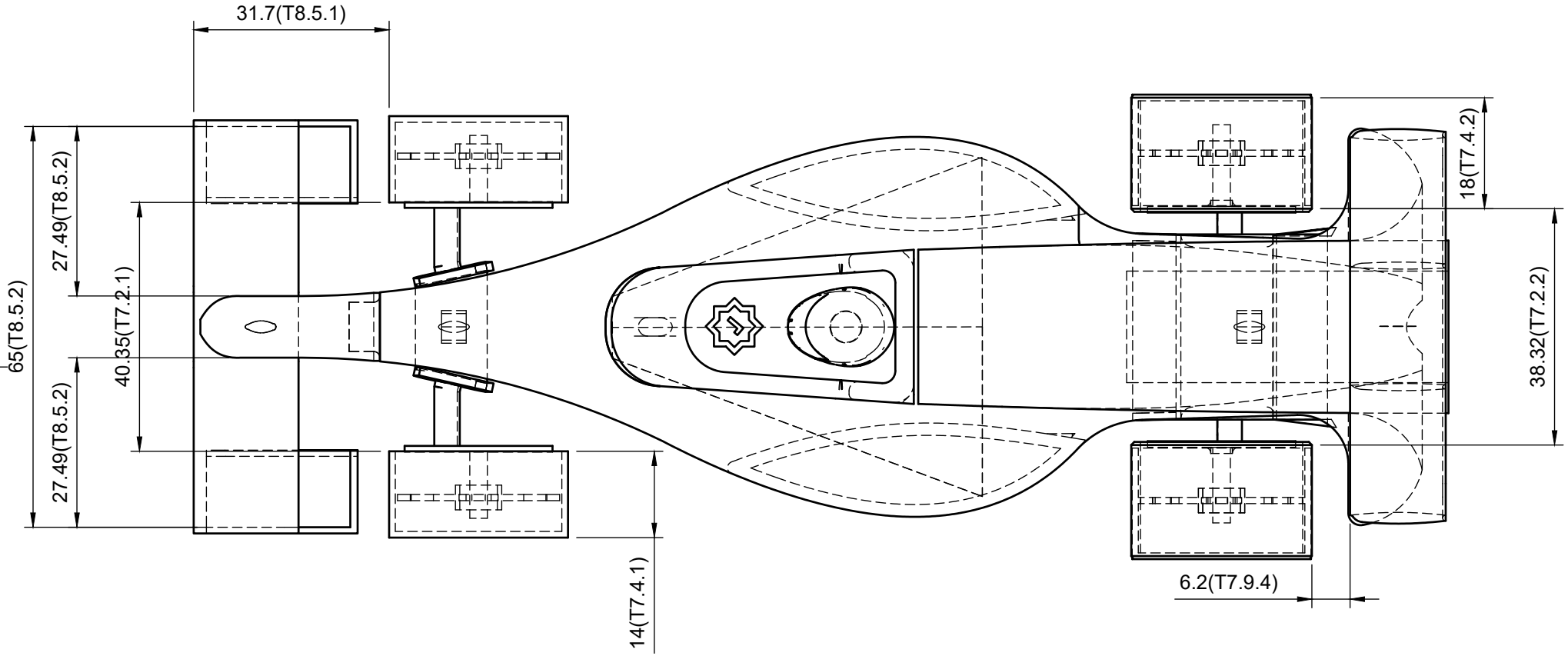




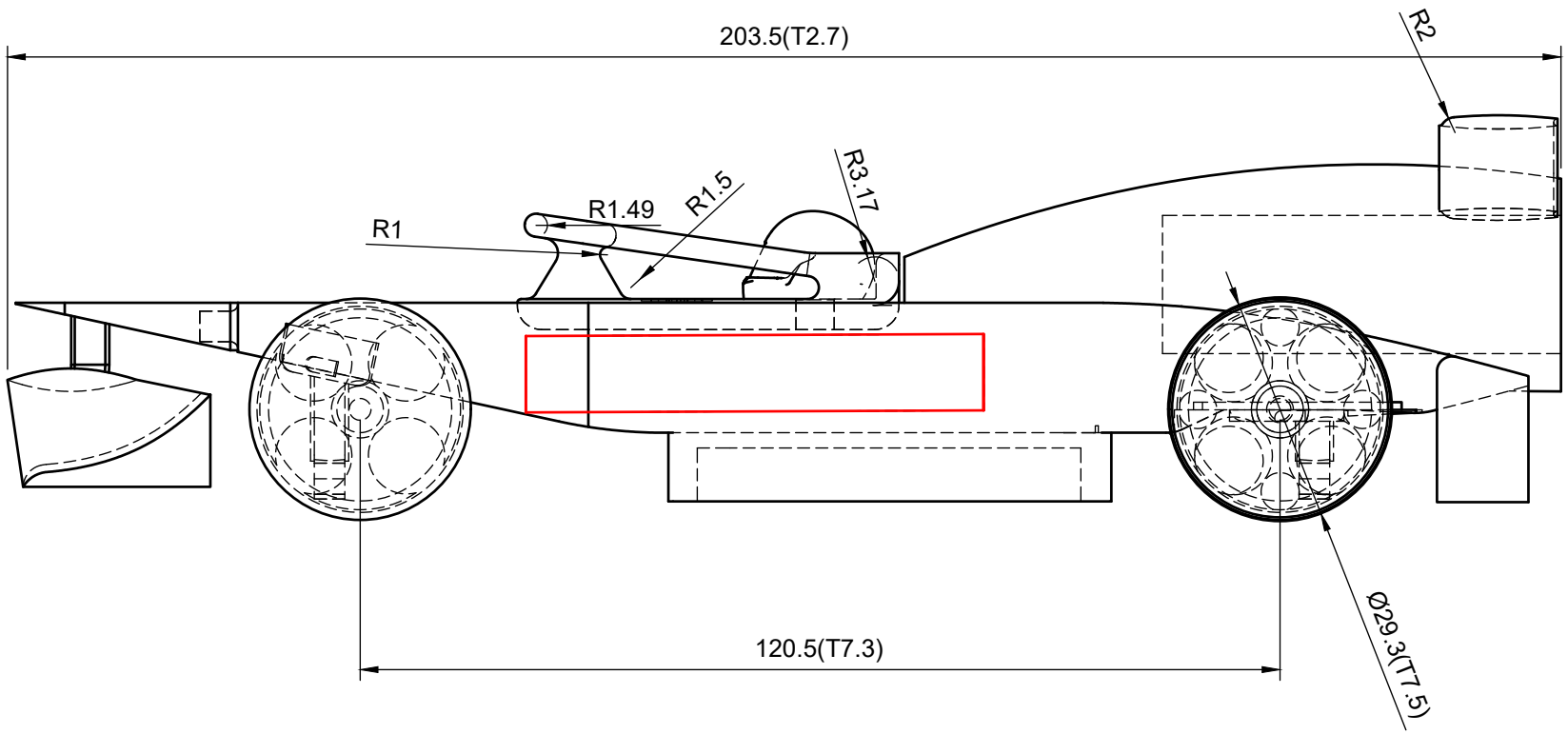



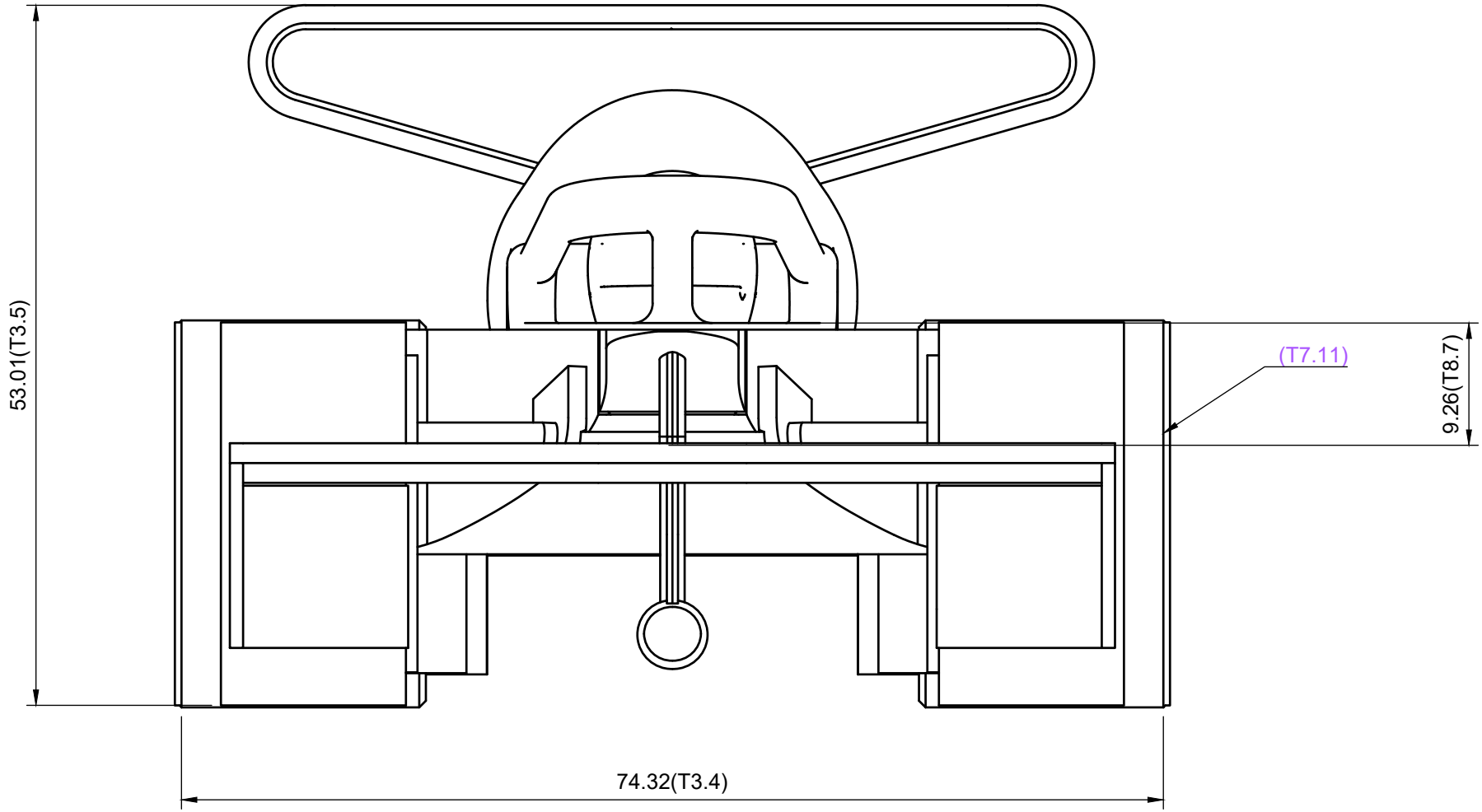

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| | T4.x | Yuvraj Nikum 02-08-2026 | Inferno Racing |
| | | Document type: Engineering Drawing | Document status: Completed |
| | | Title: f52 nationals Virtual Cargo | DWG No.: 2 |
| | | | Rev. / Date of issue: 01.08.2026 / Sheet: 2/10 |



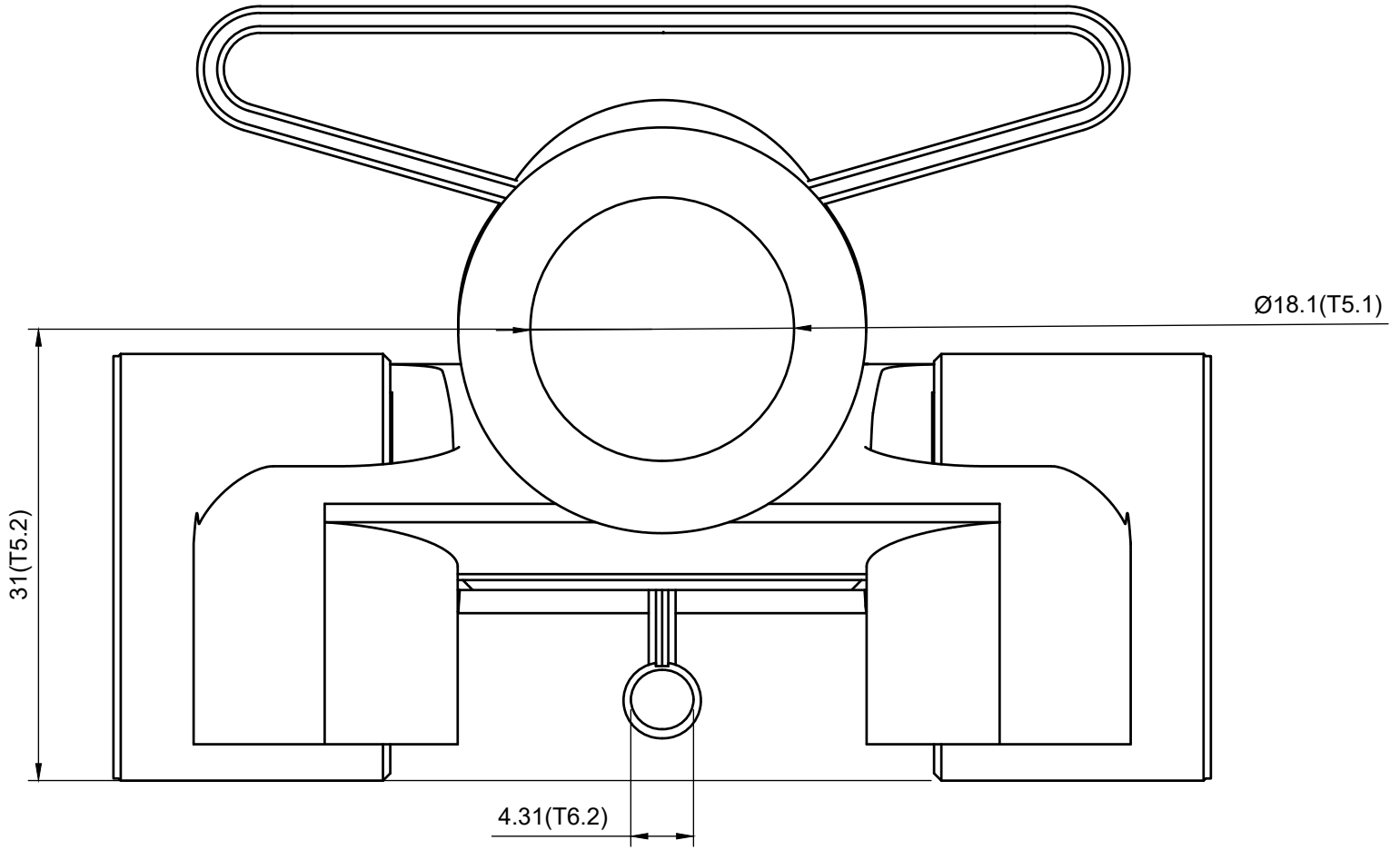

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| | | Yuvraj Nikum 02-08-2026 | Inferno Racing |
| | | Document type: Engineering Drawing | Document status: Completed |
| | | Title: f52 nationals Top View | DWG No.: 3 |
| | | | Rev. / Date of issue: 01.08.2026 / Sheet: 3/10 |



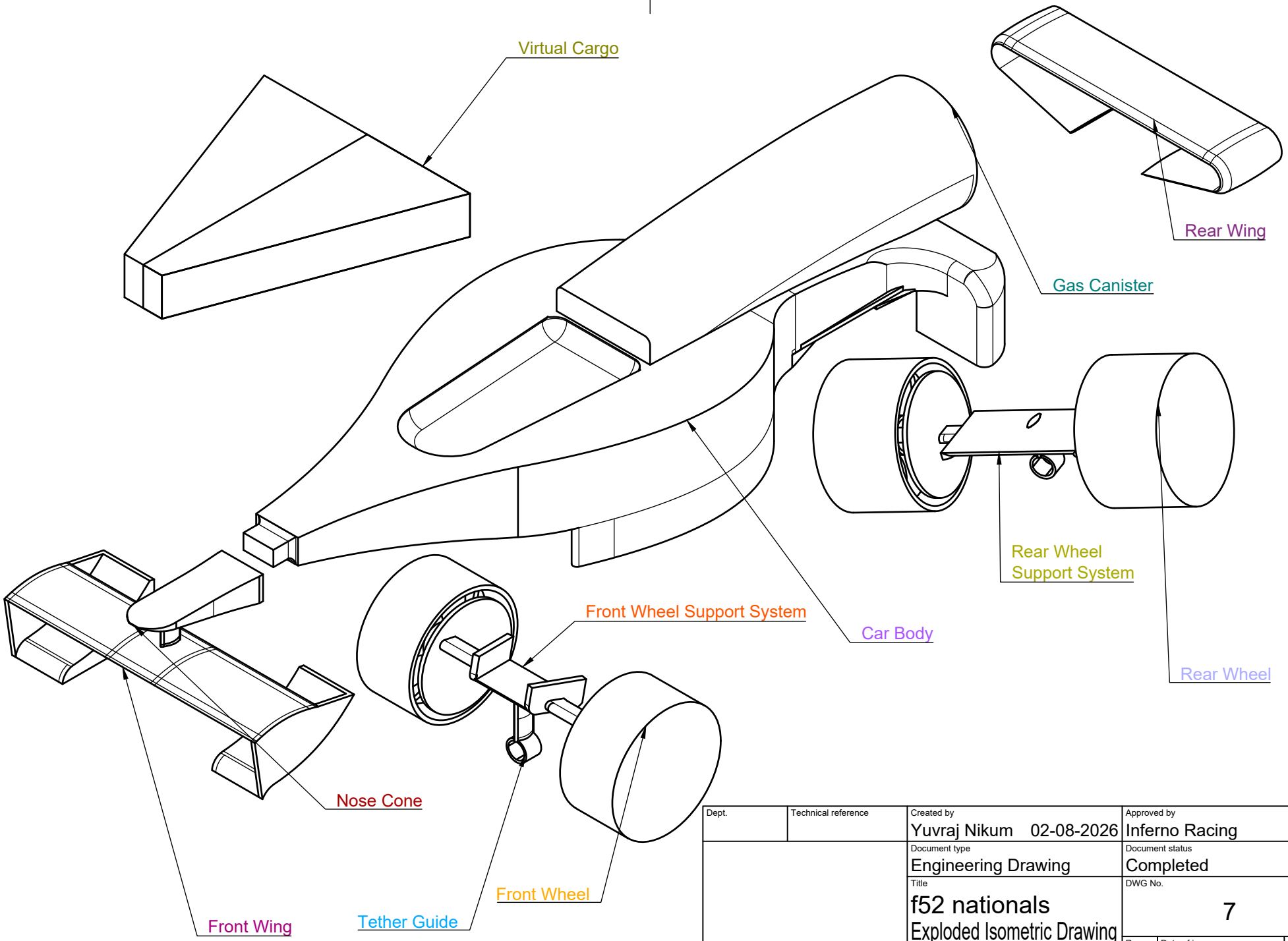

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| | | Yuvraj Nikum 02-08-2026 | Inferno Racing |
| | | Document type: Engineering Drawing | Document status: Completed |
| | | Title: f52 nationals Side View | DWG No.: 4 |
| | | | Rev. / Date of issue: 01.08.2026 / Sheet: 4/10 |




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| | | Yuvraj Nikum 02-08-2026 | Inferno Racing |
| | | Document type: Engineering Drawing | Document status: Completed |
| | | Title: f52 nationals Front View | DWG No.: 5 |
| | | | Rev. / Date of issue: 01.08.2026 / Sheet: 5/10 |




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| | | Yuvraj Nikum 02-08-2026 | Inferno Racing |
| | | Document type: Engineering Drawing | Document status: Completed |
| | | Title: f52 nationals Back View | DWG No.: 6 |
| | | | Rev. / Date of issue: 01.08.2026 / Sheet: 6/10 |




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| | | Yuvraj Nikum 02-08-2026 | Inferno Racing |
| | | Document type: Engineering Drawing | Document status: Completed |
| | | Title: f52 nationals Exploded Isometric Drawing | DWG No.: 7 |
| | | | Rev. / Date of issue: 01.08.2026 / Sheet: 7/10 |

# INTENDED QUALITY, ASSEMBLY AND FINISH — f52 nationals

**Inferno Racing · Team IR-11 · Manufacturing: Yuvraj Nikum**

## OVERALL APPROACH

Both cars are machined to identical geometry (T3.1.4) from a single STEM Racing Model Block using a Denford CNC router (T3.1.2). Our aim is two visually and dimensionally identical cars where every regulated dimension sits inside its limit with a safety margin, so machining tolerance can never push a feature out of compliance. Quality is designed in at the CAD stage and protected through careful CNC setup, minimal hand finishing, and controlled painting.

## BODY

The body is cut as one continuous piece of Model Block rear of Reference Plane A (T4.1). We machine the cartridge chamber to 52.2mm depth and Ø18.1mm, holding the 3.5mm safety zone (T5.5) around it, and keep the chamber free of paint so the diameter stays in tolerance. The intended finish is a smooth, flat body surface achieved straight off the CNC, with hand finishing (T1.13) limited to removing machining marks only, using fine abrasive paper. We keep hand finishing to a minimum so no material is unintentionally removed from regulated surfaces.

## WHEELS AND WHEEL SUPPORTS

Wheels are machined to Ø29.3mm within the 28–32mm limit (T7.5), with the track contact band kept flat and consistent (T7.7) and chamfers placed outside the contact width so they don't affect the measured surface. The intended finish is a clean, burr-free rolling surface so all four wheels contact the track evenly (T7.6). Wheel supports are finished to run freely on the axle while staying rigid for the safety test (T7.13).

## NOSE, FRONT WING AND REAR WING

Wings are finished with smooth, continuous aerofoil surfaces so the span, chord and thickness stay uniform across the full length. We aim for crisp, un-chipped leading and trailing edges and rigid wings that will not deflect during racing (T8.4, T9.3), protecting the wing-span dimensions under load. Endplates and supports are finished to blend cleanly into the wing without adding material that would break the 5mm clear-airflow zones (T8.7, T9.6).

## HALO, HELMET AND TETHER GUIDES

The Halo and Helmet are fitted as the supplied standard parts with no dimensional change (T4.4.1, T4.5). The Halo is bonded into a recess cut with a 6.35mm ball-nose cutter so the notch sits at 34.0 ±1.0mm above the track (T4.4.4). Tether guides are finished with fully closed, smooth internal holes so the tether line runs freely and passes the 200g pull test (T6.3).

## ASSEMBLY

Components are joined using adhesives with no dimensional impact (T1.1). We assemble on a flat reference surface so the car rests evenly on all four wheels with a full 8g cartridge inserted (T2.7.2). Replacement components are machined from the same CAD data and checked
to be identical to the fitted parts (T3.9.1) before submission.

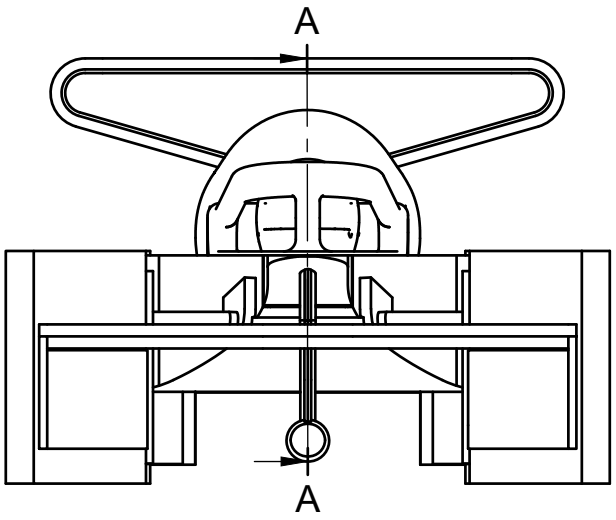
## FINISH AND IDENTITY

Painting is done in the team's identity colours with decals kept under 0.5mm thick (T4.8) and the official STEM Racing decals applied 100% visible in side view (T4.6). Decals and paint are applied after all regulated dimensions are verified, so finishing never pushes a feature out of tolerance.

## TWO IDENTICAL CARS

Car A and Car B are produced from the same CAD file and CAM program on the same machine, then measured against each other to confirm identical geometry (T3.1.4) before parc fermé.

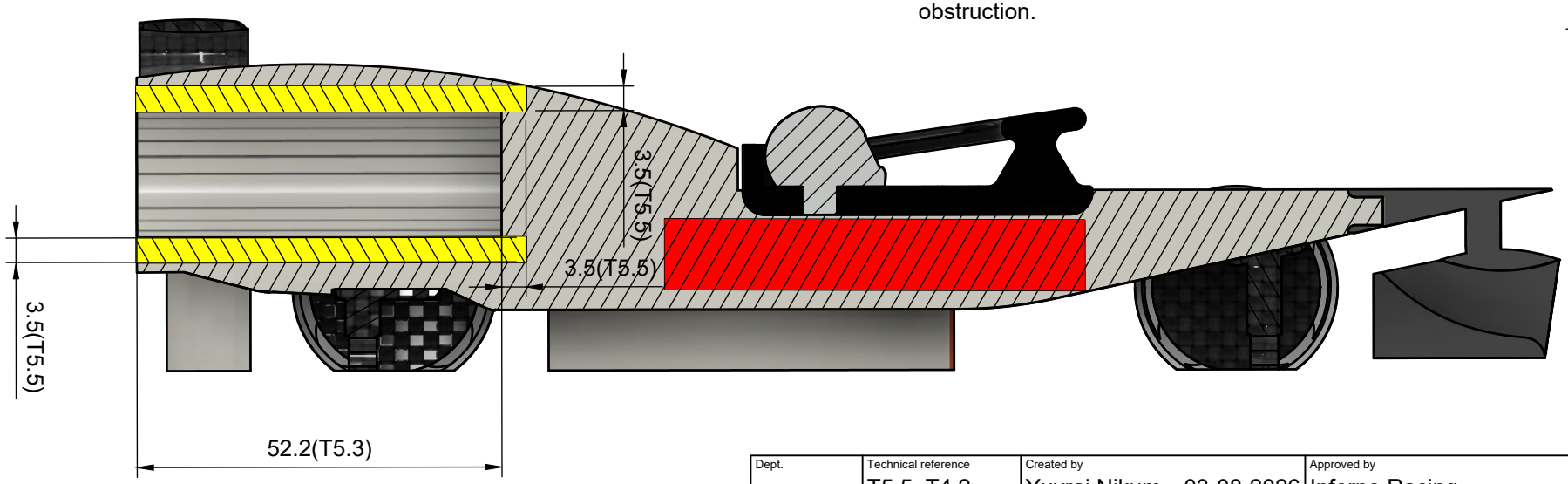
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| | | Yuvraj Nikum 02-08-2026 | Inferno Racing |
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| | | Title: f52 nationals | DWG No.: 8 |
| | | | Rev. / Date of issue: 01.08.2026 / Sheet: 8/10 |




A-A
(1.5)

## POWER UNIT CARTRIDGE

PROTRUSION – T5.6 [PERFORMANCE]
Cartridge length (App vi): 65.0mm | Chamber depth (T5.3): 52.2mm
Protrusion = 65.0 − 52.2 = 12.8mm
Absolute Min 5.0mm — COMPLIANT
Cartridge fully visible radially about its centreline — no obstruction.




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| | T5.5, T4.2 | Yuvraj Nikum 03-08-2026 | Inferno Racing |
| | | Document type: Engineering Drawing | Document status: Completed |
| | | Title: f52 nationals Chamber Safety Zone Virtual Cargo | DWG No.: 9 |
| | | | Rev. / Date of issue / Sheet: 9/10 |